

Building Bharat's own voice

INSIDE GNANI.AI'S MISSION TO CREATE INDIA'S VOICE AI FUTURE

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By all conventional wisdom, setting up a deep tech AI startup in 2017 was not the obvious career move. But for Ganesh Gopalan and Ananth Nagaraj, former colleagues at Texas Instruments, it was precisely the right kind of challenge. "We decided to set up a deep tech AI company – rather an unfashionable idea at that time," Gopalan recalls, with a wry nod to the trendlines. "Today, it's really fashionable to be in AI and do deep tech!"

The problem space they zeroed in on – Voice AI for India's multilingual, multicultural reality – was vast, untamed, and, crucially, rooted in lived experience. "Voice AI still seemed a large enough unsolved problem that could impact millions," he says. That conviction birthed Gnani.ai, a company that is now quietly powering over 10 million voice-based interactions per day across BFSI, healthcare, telecom and more.

Voice, not just text

What distinguishes Gnani isn't just its tech chops – it's the very specific choice of interface. "What began as a focus on enabling voice interfaces has expanded into developing domain-specific speech-to-speech large language models (LLMs)," Gopalan explains. These models are tuned not just to understand, but to converse – handling accents, dialects, and even code-switching across 14+ Indian languages.

This isn't some generic speech-to-text engine lazily slapped onto enterprise chatbots. It's a bespoke, culturally calibrated Voice AI stack built in and for India.

The IndiaAI leap

It's one thing to support multiple languages; it's quite another to do it in real time with 14 billion parameters under the hood. That's the ambition Gnani is now chasing under the IndiaAI Mission.

"Key technical considerations include ensuring low-latency inference, efficient memory management, and optimized compute resource usage," says Gopalan. And then there's the sheer grit of assembling diverse voice datasets, fine-tuning models to detect subtle dialectal shifts, and ensuring it all works inside regulatory frameworks. "Deploying the model responsibly requires robust infrastructure, seamless integration with enterprise systems, and adherence to data privacy and security standards," he adds.

To deliver on their 250ms response-time goal, Gnani trained its speech-to-speech LLM using over 14 million hours of proprietary multilingual data on NVIDIA's Hopper GPUs via the NeMo



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platform. "Leveraging NVIDIA's full-stack AI-accelerated computing platform – including TensorRT-LLM, Triton, and Riva – we optimized everything from voice recognition to response generation," says Gopalan.

Patents for a voice-first future

Gnani's orchestration engine also plays air traffic controller to a fleet of language models: their own SLMs, plus external ones like Google Gemini and Anthropic. With over 12 patents filed, Gopalan highlights two game-changing ones. "The first focuses on generating multilingual transcripts from multilingual audio input... we're evolving this into instant voice-to-voice translation." The second patent addresses advanced speech activity detection – critical to lesser latency and building emotionally intelligent, multilingual interactions.

Looking ahead to 2026, Gopalan's dreaming big. "The fundamental problem with most current systems is that they transcribe everything into text first," he says. Their next Voice LLM aims to decode emotion, sarcasm, and dialect-level nuance in real time. Exciting times ahead for sure. **d**